# Localizing INT8 Quantization Collapse in EfficientNet-B0 for Edge Poultry Disease Detection

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***Abstract*—We compare MobileNetV2, ShuffleNetV2, and EfficientNet-B0 for poultry-fecal disease classification on resource-constrained edge devices. The audit of the published 8,770-image benchmark finds 617 near-duplicate pairs, including 94 test images byte-identical to training images; results therefore use an 8,153-image deduplicated split. Dynamic INT8 changes MobileNetV2 and ShuffleNetV2 only slightly but collapses EfficientNet-B0 to 10.42%. Static per-channel QDQ recovers EfficientNet-B0 to 94.56%, while selective ablation indicates strong sensitivity to depthwise-layer granularity. On Raspberry Pi 5, dynamic INT8 is 1.93–2.80× slower than FP32 at four threads, whereas static per-channel QDQ reaches 94.56% at 13.50 ms, 2.34× faster than FP32. These results show that deployment outcomes depend jointly on architecture, runtime, and quantization granularity.**

***Index Terms*—deep learning, poultry disease detection, lightweight CNN, quantization, edge computing, post-training quantization**

## I. Introduction

Poultry production is a major agricultural sector in the United States (combined broiler, egg, turkey, and chicken value of $67.1 billion in 2023 [10]). Coccidiosis, Salmonellosis, and Newcastle Disease (NCD) cause major losses [2]. Fecal image analysis enables noninvasive early detection, but laboratory diagnosis remains inaccessible to many small farms [3].

Recent deep learning approaches have achieved high accuracy for poultry fecal disease classification. Dhungana *et al.* reported 99.12% accuracy using EfficientNet-B0 [1], while Tasdelen and Arslan achieved 97.1% with MobileNetV2 [3]. However, these solutions primarily target cloud-based deployment, leaving a critical gap: *how can these models be efficiently deployed on resource-constrained edge devices at farm level?*

Lightweight CNNs—MobileNetV2 [5], ShuffleNetV2 [6], and EfficientNet [7]—are candidates for edge deployment because they need fewer operations. INT8 post-training quantization (PTQ) reduces size by using 8-bit weights [8]. Even on CPUs that expose INT8 acceleration, PTQ can increase latency rather than reduce it when the runtime kernel path adds quantize/dequantize work around every convolution; our x86 reference (12th Gen i5-12400F, AVX-VNNI present per `/proc/cpuinfo`) still shows a 3.0–16.8× slowdown, so the slowdown is consistent with the ORT CPU kernel path adding quantize/dequantize work, not raw ISA absence; kernel-level attribution is discussed cautiously in Sec. IV-C. See also the quantization white paper for why PTQ without calibration degrades on depthwise layers [14].

This paper addresses two research questions: (1) What is the quantization sensitivity of each lightweight architecture under INT8 PTQ on CPU, and where exactly does EfficientNet-B0 fail? (2) What is the accuracy-latency tradeoff for these models under INT8 PTQ on a Raspberry Pi 5? We additionally audit the dataset integrity claim underlying prior work on this benchmark.

The main contributions of this paper are summarized as follows. First, we audit the FecalFed leak-free claim [9] and identify 617 near-duplicate pairs, including 393 byte-exact pairs and 94 test images byte-identical to training images; all results use the resulting 8,153-image deduplicated split with 5,100/1,325/1,728 train/validation/test images and zero cross-split survivors. Second, we characterize EfficientNet-B0 sensitivity to quantization granularity: dynamic INT8 and static per-tensor QDQ collapse to 10.42% and 9.32%, respectively, while static per-channel QDQ recovers 94.56%. Selective ablations and layerwise SQNR provide evidence that depthwise layers are especially sensitive in this configuration, but do not establish a universal necessary-and-sufficient mechanism. Third, we characterize the three models on Raspberry Pi 5 across 12 model–quantization–thread configurations and compare dynamic with static QDQ, showing that deployment latency depends on the runtime and quantization scheme.

## II. Related Work

### A. Poultry Disease Detection

Machuve *et al.* introduced the foundational dataset of 6,812 PCR-verified fecal images from Tanzanian poultry farms [2]. Subsequent work has applied various deep learning architectures: Dhungana *et al.* combined YOLOv11 detection with EfficientNet-B0 classification, achieving 99.12% accuracy and

25.8 ms inference [1]. Tasdelen and Arslan benchmarked six architectures, finding MobileNetV2 optimal for both accuracy (97.1%) and training speed [3]. Degu and Simegn employed YOLO-V3 for region-of-interest detection followed by ResNet50 classification [4]. Chi introduced FecalFed, a federated-learning benchmark on this dataset family claiming 8,770 rigorously deduplicated unique images [9]; our audit in Section III-A tests that claim directly. Prior work on this dataset has not compared lightweight models under quantization with accuracy-latency tradeoff analysis.

### B. Lightweight CNN Architectures

MobileNetV2 [5] utilizes inverted residuals with linear bottlenecks and depthwise separable convolutions for efficient feature extraction. ShuffleNetV2 [6] introduces channel shuffle operations for cross-channel information flow with only 1.26M parameters. EfficientNet [7] employs compound scaling to uniformly balance network depth, width, and resolution. Hoang *et al.* recently compared seven lightweight architectures for plant disease detection, identifying hybrid designs as optimal for edge deployment [11].

### C. Model Quantization

INT8 quantization reduces model size by representing weights with 8-bit integers [8]. While theoretically achieving $4\times$ compression, ONNX Runtime documentation [12] explicitly notes that INT8 without matching kernel support may degrade performance due to quantize/dequantize overhead. The white paper on quantization surveys why PTQ without calibration degrades on depthwise and SE layers [14].

## III. Methodology

### A. Dataset

We use the Machuve *et al.* dataset [13] with PCR-verified fecal images from Tanzanian farms, as distributed via HuggingFace `Dianyo/poultry-fecal-fl`. Chi [9] describes this benchmark as 8,770 rigorously deduplicated leak-free images with train/test splits of 7,016/1,754. We audit this claim with pHash (Hamming $\leq 5$) over all 8,770 files and find 617 near-duplicate pairs (393 byte-exact), including 303 groups spanning splits: 94 test images byte-identical to train images and 31 to validation images.

All numbers in this paper are therefore reported on a deduplicated sterile split: we drop one copy per duplicate pair (keeping test $>$ val $>$ train), yielding 8,153 unique images with 5,100/1,325/1,728 train/val/test and zero cross-split survivors under this audit (audit artifact `results/phash_dedup.json`, split manifest `results/sterile_split_manifest.json`).
Concretely: the audit found 617 near-duplicate pairs (all size 2, class-consistent; 393 byte-exact), of which 303 groups span splits; we drop exactly one copy per pair keeping test $>$ val $>$ train (within a split, lexicographically-first kept), so counts refer to removed copies rather than connected groups. The validation split was carved as a stratified 80/20 split of the publisher train only (seed 42); publisher test was never re-split. We use the shorthand *sterile split* throughout for this pHash-deduplicated partition. The term denotes only the absence of cross-split near-duplicate pairs detected by the audit described above; it is not a claim of independence between images, since residual dependence may remain within a split. We note as a limitation that this removes leakage but does not rebuild a fresh split from the Zenodo originals (6,812 images); the sterile test set remains 98% inherited from the publisher test split. Classes are Coccidiosis, Healthy, Salmonella, and Newcastle Disease; NCD is underrepresented (417/5,100 train, 141/1,728 test), so we report per-class INT8 recall: Coccidiosis 2.27%, Healthy 4.76%, Salmonella 0.54%, and Newcastle Disease 100%. Images are resized to $224\times224$ and normalized with ImageNet mean/std.

### B. Model Architectures

We evaluate three ImageNet [15]-pretrained lightweight architectures with the classification head replaced for four classes (3.4M$\rightarrow$2.2M for MobileNetV2 after the head swap; parameter counts below are for our 4-class models). MobileNetV2 [5] (2.2M parameters) uses inverted residuals with depthwise separable convolutions for efficient local feature extraction. ShuffleNetV2 [6] (1.3M parameters) relies on channel shuffle for cross-channel mixing with few parameters. EfficientNet-B0 [7] (4.0M parameters) applies compound scaling with squeeze-and-excitation blocks to capture multi-scale features.

### C. Training Protocol

All models are ImageNet-pretrained (torchvision V1 weights) with only the classification head replaced (4 classes); all layers are trainable (no freezing). Each model is trained once (seed 42) for 50 epochs using Adam (learning rate $10^{-4}$, weight decay $10^{-4}$) with cosine annealing; the checkpoint with the best validation accuracy is kept. Batch size is 32, input resolution $224 \times 224$. Data augmentation (train only): random horizontal flip, rotation ($\pm15°$), color jitter (brightness/contrast/saturation 0.2); eval preprocessing is resize$\rightarrow$tensor$\rightarrow$ImageNet normalize. Training is performed on an NVIDIA RTX 4060 GPU with 8 GB VRAM using PyTorch 2.5.1 with CUDA 12.1.

### D. Quantization and Benchmarking (x86)

Models are exported to ONNX (opset 13) and quantized with ONNX Runtime dynamic per-tensor INT8 using signed QInt8 weights and dynamically quantized activations; default operator selection includes convolution operators observed in the resulting graphs. The original static-QDQ run used the first 100 training images sequentially; because the dataset loader is class-ordered, this calibration set contained only Coccidiosis images and was not class-balanced. We therefore treat the static-QDQ accuracy as a single-slice diagnostic result and list stratified recalibration as required follow-up, not as a claim of a generally calibrated deployment. Static QDQ uses signed QInt8 weights and activations, MinMax calibration, per-tensor or per-channel weight scales, and standard ORT quantization

preprocessing; no custom graph fusion is enabled. The native-Linux rerun, performed with the same ORT model-only protocol, gives dynamic-INT8 slowdowns of 14.73–14.97× for MobileNetV2, 3.22–3.26× for ShuffleNetV2, and 14.08–14.08× for EfficientNet-B0 at one and four threads, respectively. These host-reference values are reported separately from the WSL and Raspberry Pi 5 results.

### *E. Edge Deployment on Raspberry Pi 5*

We characterize the same ONNX FP32 and INT8 models on a Raspberry Pi 5 (Broadcom BCM2712, 4× Cortex-A76 at up to 2.4 GHz, 8 GB LPDDR4X, Debian 13 trixie, kernel 6.18.34+rpt-rpi-2712, aarch64, `ondemand` governor, throttled=0x0). ONNX Runtime 1.29.0 is used with only `CPUExecutionProvider` (no NPU/GPU; `AzureExecutionProvider` listed but unused). The sterile-model sweep (`scripts/benchmark_pi_gate0.py`) covers 12 configurations: 3 models × 2 quantizations (FP32/INT8 dynamic) × 2 thread counts (1 and 4 via `intra_op_num_threads`) × batch 1, with input 3×224×224. Each configuration runs 200 iterations after 10 warmup iterations (single session per config), reporting mean/std/P50/P95/P99 latency, throughput, RSS memory (mean/peak), CPU%, and die temperature before/after. Variability reported is within-session; variability across independent benchmark sessions was not measured. All timings are model-only (eval preprocessing and data transfer excluded), not end-to-end application throughput. Initial temperature was 44.6°C; peak during the sweep was 55.6°C. Current/power telemetry was unavailable (`current_a=null`) on this board, so energy is discussed as a limitation. All 12/12 runs completed without thermal throttling.

## IV. Results and Discussion

### *A. Classification Performance*

Table I summarizes the classification performance on the sterile test set (1,728 images). All three architectures achieve over 97% accuracy, demonstrating the effectiveness of ImageNet transfer learning for the poultry fecal domain. Dynamic INT8 reduces accuracy slightly for MobileNetV2 and ShuffleNetV2; the paired test detects discordance for MobileNetV2 ($b$=18, $c$=7, $p$=0.0455) but not for ShuffleNetV2 ($b$=6, $c$=5, $p$=1.000). This test does not establish equivalence. The EfficientNet-B0 collapse is 88.13 percentage points.

TABLE I
TEST ACCURACY ON STERILE SPLIT (1,728 IMAGES); DYNAMIC INT8 PTQ (ORT). MACRO COLUMNS ARE MACRO-AVERAGED PRECISION/RECALL/F1 AND BALANCED ACCURACY.

| Model | FP32 acc (%) | INT8 acc (%) | Drop (pp) | FP32 macro-F1 (bal. acc) | INT8 macro-F1 (bal. acc) |
|---|---|---|---|---|---|
| MobileNetV2 | 97.92 | 97.28 | 0.64 | 97.57 (97.26) | 96.45 (96.09) |
| ShuffleNetV2 | 97.69 | 97.63 | 0.06 | 97.10 (97.34) | 96.82 (96.76) |
| EfficientNet-B0 | 98.55 [97.97, 99.07] | 10.42 [8.97, 11.92] | 88.13 | 98.22 (98.03) | 7.49 (26.89) |

Table II reports the EfficientNet-B0 static-QDQ comparison. Table III summarizes the selective per-family ablation: depthwise-only per-channel scaling reaches 93.46%, while excluding depthwise from per-channel scaling yields 8.33%. The exported-weight audit in Table V reports median P95/P5 depthwise range ratios of 39.94× for EfficientNet-B0, 9.20× for MobileNetV2, and 5.32× for ShuffleNetV2; Table IV reports the EfficientNet-B0 family-level SQNR values.

TABLE II
EFFICIENTNET-B0 STATIC QDQ INT8 DIAGNOSTIC USING MINMAX CALIBRATION ON THE FIRST 100 CLASS-ORDERED TRAINING IMAGES (COCCIDIOSIS ONLY). PER-TENSOR REPRODUCES THE COLLAPSE; PER-CHANNEL RECOVERS. MCNEMAR TESTS USE CONTINUITY CORRECTION; $b$ AND $c$ ARE DISCORDANT-PAIR COUNTS.

| Config | Test acc. (%) | vs FP32 $b/c$ | vs FP32 $p$ | vs dynamic $b/c$ | vs dynamic $p$ |
|---|---|---|---|---|---|
| Dynamic INT8 | 10.42 | — | — | — | — |
| Static QDQ per-tensor | 9.32 | 1548/7 | $< 10^{-15}$ | 36/17 | 0.013 |
| Static QDQ per-channel | 94.56 | 75/7 | $1.37 \times 10^{-13}$ | 14/1468 | $< 10^{-15}$ |

TABLE III
SELECTIVE PER-FAMILY STATIC QDQ ABLATION FOR EFFICIENTNET-B0 (MINMAX, FIRST 100 CLASS-ORDERED TRAINING IMAGES, COCCIDIOSIS ONLY). FAMILIES ARE GROUPED FROM THE ONNX GRAPH; FP32 REFERENCE ACCURACY IS 98.50%.

| Config. | | Size | Acc. | Collapse |
|---|---|---|---|---|
| A | All per-tensor | 4.29 | 9.32 | yes |
| B | All per-channel | 4.67 | 94.56 | no |
| C | **Depthwise** per-channel, rest per-tensor | 4.33 | **93.46** | no |
| D | All per-channel *except* depthwise | 4.51 | 8.33 | yes |
| E | Depthwise left in FP32 | 4.69 | 97.16 | no |
| F | Squeeze-and-excitation left in FP32 | 5.91 | 12.44 | yes |

TABLE IV
LAYERWISE WEIGHT SQNR UNDER SIMULATED INT8. VALUES ARE FAMILY MEANS; GAIN IS THE PER-CHANNEL IMPROVEMENT.

| Family | Tensors | Params | Per-tensor | Per-channel (gain) |
|---|---|---|---|---|
| **Depthwise** | 16 | 182k | **27.97** | 44.42 (+16.46) |
| SE reduce | 16 | 314k | 32.68 | 37.14 (+4.46) |
| SE expand | 16 | 314k | 34.00 | 45.26 (+11.26) |
| Pointwise | 32 | 3.15M | 33.67 | 42.30 (+8.63) |
| Stem | 1 | 864 | 35.19 | 45.28 (+10.09) |

TABLE V
DEPTHWISE WEIGHT AUDIT ACROSS ARCHITECTURES. RATIOS ARE MEDIAN PER-LAYER P95/P5 CHANNEL RANGES; SQNR IS MEDIAN ACROSS DEPTHWISE LAYERS.

| Model | DW layers | Range P95/P5 | Median SQNR (dB) |
|---|---|---|---|
| EfficientNet-B0 | 16 | 39.94 | 16.67 |
| MobileNetV2 | 17 | 9.20 | 34.06 |
| ShuffleNetV2 | 19 | 5.32 | 37.79 |

EfficientNet-B0 has the highest FP32 accuracy (98.55%), but dynamic INT8 collapses to 10.42%; Fig. 1 shows that 1,676/1,728 predictions fall in the Newcastle Disease column. Static per-tensor QDQ also collapses to 9.32%, whereas

per-channel QDQ recovers 94.56% (Table II). A plausible mechanism is channel-wise range imbalance: the shared per-tensor scale is set by the largest channel, reducing resolution for smaller channels. EfficientNet-B0 depthwise layers reach 1,152 channels, and the audit reports lower depthwise SQNR than the corresponding MobileNetV2 and ShuffleNetV2 audit. These observations support sensitivity to granularity but do not prove a universal mechanism; the experiment uses one seed and one calibration set. Per-class accuracy denotes recall; macro metrics for all models are reported in Table I.

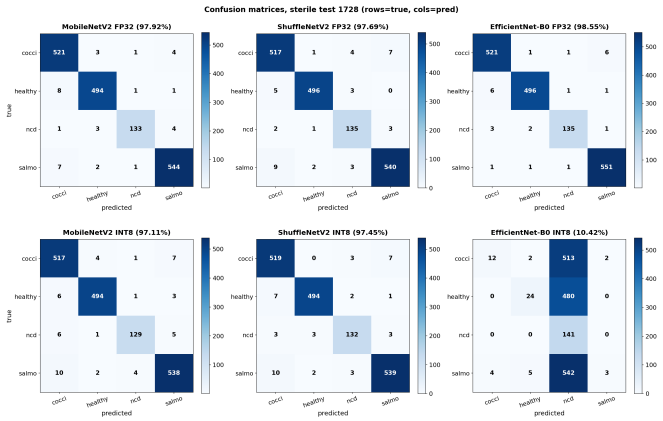



Fig. 1. Confusion matrices on the sterile test set (1,728 images). EfficientNet-B0 changes from high-quality FP32 predictions to near-mode collapse under dynamic INT8, with most predictions assigned to Newcastle Disease.

### B. Quantization Impact

Table VI presents the quantization analysis on the sterile split. All architectures achieve 3.3–3.7× model size reduction through INT8 quantization.

TABLE VI
INT8 QUANTIZATION ANALYSIS (WSL X86 REFERENCE, BATCH 1, 200 RUNS + 10 WARMUP, MODEL-ONLY TIMING, DEFAULT THREADS). STERILE-SPLIT MODELS.

| Model | Size | | Latency | | Ratio |
|---|---|---|---|---|---|
| | FP32 | INT8 | FP32 | INT8 | INT8/FP32 |
| MobileNetV2 | 8.48 | 2.30 | 2.40 | 40.25 | 16.8× |
| ShuffleNetV2 | 4.89 | **1.47** | 4.15 | 12.62 | 3.0× |
| EfficientNet-B0 | 15.3 | 4.16 | 4.35 | 57.18 | 13.1× |

INT8 quantization increases latency by 3.0–16.8× versus FP32 on the x86 reference. Our reference CPU exposes AVX-VNNI yet still slows down, suggesting the ORT CPU kernel path (quantize/dequantize ops around the matmul) dominates; graph operators alone do not prove which kernels execute, so Sec. IV-C attributes this cautiously using executed-kernel profiling on RPi5.

ShuffleNetV2 degrades least (3.0× versus 16.8× for MobileNetV2). Channel-shuffle ops are non-parametric and less sensitive to precision, while MobileNetV2 depthwise separable convolutions are more sensitive.

### C. Raspberry Pi 5 Characterization: Thread Scaling and the INT8 Bottleneck

Table VII and Table VIII summarize the sterile-model Raspberry Pi 5 latency at batch=1. Fig. 2 visualizes the thread-scaling contrast. FP32 scales with threads: MobileNetV2 drops from 33.16 ms (1 thread, 30.2 FPS) to 14.23 ms (4 threads, 70.3 FPS, 2.33× speedup); ShuffleNetV2 from 14.19 ms to 6.19 ms (2.29×, 161.5 FPS); EfficientNet-B0 from 69.45 ms to 31.56 ms (2.20×). The sweep covers threads 1 and 4; intermediate thread counts are not measured for the sterile models.

The dynamic INT8 slowdown reported in Table VIII is a property of dynamic quantization on this runtime, not of INT8 in general. Table VII repeats the EfficientNet-B0 measurement on the same board, same ORT build, and same session with the static QDQ graphs whose accuracy we characterized in Table II. Static QDQ is 2.31–2.34× *faster* than FP32 at 4 threads (13.63 and 13.50 ms versus 31.53 ms), reversing the direction of the dynamic result (63.99 ms, 0.49× of FP32). The static per-channel result is faster and more accurate in this experiment, but it does not establish that granularity always dominates bit width: the comparison uses one board, one runtime, one seed, and one calibration set, and leaving depthwise layers in FP32 reaches 97.16%.

TABLE VII
EFFICIENTNET-B0 ON RASPBERRY PI 5: DYNAMIC VS STATIC QDQ INT8, BATCH=1, MEAN OF 200 RUNS AFTER 10 WARMUP, SINGLE SESSION PER CONFIGURATION, CPUEXECUTIONPROVIDER ONLY, ORT 1.29.0, PEAK 49.4°C, THROTTLED=0X0. ACCURACY IN THE LAST COLUMN IS MEASURED ON THE EXPORTED ONNX GRAPHS RATHER THAN THE PYTORCH CHECKPOINTS REPORTED IN TABLE I, WHICH IS WHY THE FP32 REFERENCE READS 98.50% HERE VERSUS 98.55% THERE; THE 0.05-POINT GAP IS ONE IMAGE AND REFLECTS FLOAT32 EXPORT. PER-CHANNEL STATIC QDQ IS BOTH THE ACCURATE AND THE FAST CONFIGURATION.

| Quantization | Granularity | Size (MB) | t=1 (ms) | t=4 (ms) | Rel. FP32 t=4 | Acc. (%) |
|---|---|---|---|---|---|---|
| FP32 | — | 15.30 | 69.00 (68.97) | 31.53 (30.89) | 1.00× | 98.50 |
| INT8 dynamic | per-tensor | 4.16 | 76.38 (76.35) | 63.99 (63.82) | 0.49× | 10.42 |
| INT8 static | per-tensor | 4.29 | 24.51 (24.50) | 13.63 (13.37) | 2.31× | 9.32 |
| INT8 static | **per-channel** | 4.67 | 24.66 (24.64) | **13.50** (13.49) | 2.34× | 94.56 |

TABLE VIII
RASPBERRY PI 5 LATENCY, BATCH=1 (MEAN OF 200 RUNS, P50 IN PARENTHESES). STERILE-SPLIT MODELS, ORT 1.29.0, 44–56°C, THROTTLED=0X0.

| Model | Quant | t=1 (ms) | t=4 (ms) | Scaling 1→4 | FP32→INT8 t=4 |
|---|---|---|---|---|---|
| MobileNetV2 | FP32 | 33.16 (33.00) | **14.23** (14.03) | 2.33× | — |
| | INT8 | 43.88 (43.86) | 39.83 (39.73) | 1.10× | 2.80× slower |
| ShuffleNetV2 | FP32 | 14.19 (14.19) | **6.19** (6.18) | 2.29× | — |
| | INT8 | 13.65 (13.64) | 11.95 (11.76) | 1.14× | 1.93× slower |
| EfficientNet-B0 | FP32 | 69.45 (69.42) | **31.56** (31.18) | 2.20× | — |
| | INT8 | 76.68 (76.67) | 64.39 (64.19) | 1.19× | 2.04× slower |

Dynamic INT8 is slower than FP32 at four threads (1.93–2.80×), and its thread scaling is limited: from 1 to 4 threads, INT8 improves only 1.10–1.19× versus 2.20–2.33× for FP32. This indicates limited parallelization of INT8 kernels in

ONNX Runtime 1.29.0 on Cortex-A76. One exception is ShuffleNetV2 INT8 at one thread (13.65 ms), which is slightly faster than its FP32 counterpart (14.19 ms).

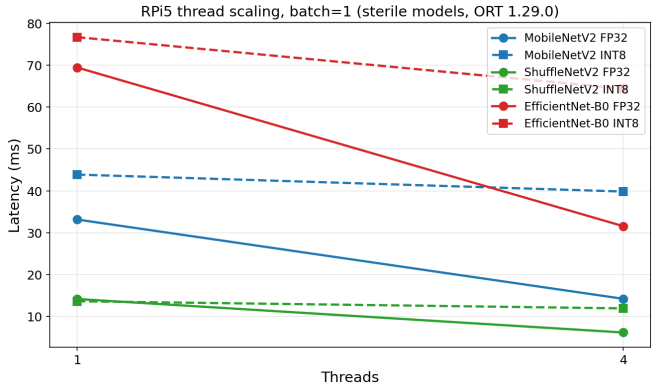



Fig. 2. Raspberry Pi 5 thread scaling at batch 1. FP32 benefits more from four threads, whereas dynamic INT8 shows limited scaling and remains slower at four threads.

The exported-weight audit also records depthwise channel range imbalance across architectures in `results/depthwise_range_audit.json`; we use it as diagnostic evidence rather than a causal claim.

Model file size shrinks 3.3–3.7× with INT8 (e.g., 8.48→2.30 MB for MobileNetV2). On the sterile RPi5 sweep (batch=1 only), runtime RSS is similar across quantizations for MobileNetV2 (∼99–108 MB) and ShuffleNetV2 (∼111–112 MB), so INT8 does not save RAM at inference on this stack—a practical caveat for memory claims. We do not claim batch-scaling behavior for the sterile models; only batch=1 was measured.

TABLE IX
LEVEL-2 PROVENANCE: NODE EXPANSION AND EXECUTED-KERNEL DISPATCH ON RPI5. DYNAMIC INT8 DISPATCHES CONVINTEGER AND QUANTIZATION-RELATED OPERATORS; FP32 HAS NO QUANTIZATION KERNELS. WITH THREE ARCHITECTURES, THE MEASUREMENTS SUPPORT THE OBSERVED SLOWDOWN RANKING BUT NOT A GENERAL EXPANSION CORRELATION.

| Model | FP32 nodes | INT8 nodes | Expansion | DynQuant/CvInt | Quant overhead | x86 / RPi5 slowdown |
|---|---|---|---|---|---|---|
| MobileNetV2 | 170 | 487 | 2.86× | 53 / 52 | 29.2% | 16.8× / 2.80× |
| ShuffleNetV2 | 902 | 1240 | 1.37× | 54 / 56 | 20.7% | 3.0× / 1.93× |
| EfficientNet-B0 | 239 | 730 | 3.05× | 82 / 81 | 24.9% | 13.1× / 2.04× |

**Measured bottleneck (Level-2).** Table IX quantifies why INT8 is slower. Dynamic PTQ does not insert static QuantizeLinear/DequantizeLinear nodes; it expands each Conv (52–81) into a DynamicQuantizeLinear + ConvInteger + Mul (scale) + Cast pattern, so node count grows by 1.37–3.05× on the sterile graphs. ORT kernel profiling on RPi5 (enable_profiling, 15+5 runs) shows dispatched kernels: FP32 has zero quantization kernels, while INT8 dynamic dispatches ConvInteger 51–58%, DynamicQuantizeLinear 9–15%, and Mul/Add/Cast 20–35% combined; all quantization-related kernels add 20.7–29.2% overhead. The slowdown ranking agrees with the measured overhead ranking for these three models, but the sample is too small to claim a general expansion-overhead correlation. CPU feature detection exposes ARM dot-product support (`asimddp`) but not `i8mm` or SVE; whether the MLAS kernels use the dot-product instructions was not verified. Artifact `kernel_dispatch_artifact.json` contains the per-operator timings and operation counts, complementing TinyML system studies [16].

### D. Accuracy-Latency Tradeoff

The accuracy-latency comparison shows that ShuffleNetV2 FP32 is fastest at 6.19 ms on RPi5 (161.5 FPS), EfficientNet-B0 is most accurate at 98.55%, and MobileNetV2 FP32 is intermediate (97.92%, 14.23 ms). The gap between best and worst FP32 accuracy is only 0.86pp (97.69% versus 98.55%); hardware constraints and quantization scheme should drive model choice for deployment. By "balanced" we mean MobileNetV2 FP32 sits between the latency extremes (14.23 ms vs 6.19 ms ShuffleNetV2 and 31.56 ms EfficientNet-B0 at 4 threads) while staying within 0.63pp of the best accuracy. All latency numbers are means of 200 runs after 10 warmup iterations (batch 1, model-only). On x86, ShuffleNetV2 FP32 (4.15 ms) is slower than MobileNetV2 FP32 (2.40 ms) but the order flips on RPi5 (6.19 ms vs 14.23 ms at 4 threads); reference and deployment hardware rank architectures differently, which is why deployment claims rest on RPi5 only.

### E. Comparison with Related Work

Our accuracy (98.55%) is competitive with prior results on the same dataset family, although the protocols differ in splits, deduplication, and metrics, so this is not a head-to-head benchmark. Our added analysis in quantization diagnosis and accuracy-latency tradeoff characterization under INT8 is absent from prior work.

## V. CONCLUSION

This paper characterizes lightweight CNN architectures with INT8 quantization for poultry fecal disease detection. FP32 accuracy exceeds 97% for all models; dynamic INT8 yields 97.28% for MobileNetV2 and 97.63% for ShuffleNetV2, but collapses EfficientNet-B0 to 10.42%. Static per-channel QDQ recovers EfficientNet-B0 to 94.56%, while the EfficientNet-B0 ablation indicates strong sensitivity to depthwise-layer granularity (93.46% when depthwise layers alone are per-channel versus 8.33% when they are not). Dynamic INT8 is 3.0–16.8× slower on x86 and 1.93–2.80× slower on RPi5 at four threads, whereas static per-channel QDQ is 2.34× faster than FP32 on RPi5. ShuffleNetV2 FP32 is fastest on RPi5 at 6.19 ms, while MobileNetV2 FP32 provides an intermediate latency of 14.23 ms with 97.92% accuracy.

This paper characterizes one board, one runtime build, one dataset, one random seed, and one 100-image calibration set; multi-seed confidence intervals, stratified recalibration, and calibration-size sweeps are not included in the reported tables. Energy measurement was unavailable, and the RPi5 latency measurements were model-only at batch 1. The selective quantization ablation was performed only for EfficientNet-B0; extending it to MobileNetV2 and ShuffleNetV2 would

test whether the observed depthwise sensitivity generalizes. Quantization-aware training and class-imbalance mitigation for the underrepresented NCD class also remain open.

Code and sterile-split artifacts (dedup audit, split manifests, eval JSONs, benchmark logs, plotting scripts) are available at https://github.com/jaweed3/paper_poultry_p2/tree/post30-gate0-dedup.

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